

SATHANA S

AI & Data Science Graduate

[Email](#) | [Phone](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

AI & Data Science graduate skilled in Python, SQL, OOP, and Data Structures, with an IEEE-published research paper on AI-driven surveillance systems. Passionate about software development and problem-solving, seeking opportunities as a Python Developer, Software Engineer, or Graduate Engineer Trainee.

SKILLS:

TECHNICAL SKILLS

Programming: Python

Core Concepts: Object-Oriented Programming (OOP), Data Structures, File Handling, Exception Handling

Database: SQL, DBMS

Software Development: Problem Solving, Debugging, Algorithm Design

Tools: Git, GitHub, VS Code, Jupyter Notebook

RESEARCH & PUBLICATION:

1. "An Intelligent Surveillance System for Smoking-Related Risk Detection in Campus Zones" | IEEE ICSSAS-2026
 - Developed an intelligent surveillance system to automate smoking and fire-related risk detection using AI-based video analytics.
 - Implemented a multi-step verification pipeline (combining smoke pattern detection and behavioral analysis) to reduce false alarms, achieving 99.87% accuracy.
 - Engineered a risk-level classification module (Low/Medium/High) to prioritize security interventions and minimize storage by logging only verified incident clips.

PROJECTS:

1. **Online Test Portal**
 - Developed a Python-based online test portal with quiz and score evaluation features.
 - Implemented automated result generation for user assessments.
2. **Student Management System**
 - Built a student record management system to add, update, search, and delete student details.
 - Used file handling techniques for data storage and retrieval.
3. **ATM Management System**
 - Developed a menu-driven ATM application with deposit, withdrawal, and balance enquiry features.
 - Implemented transaction validation and error handling for smooth operations.

EDUCATION:

- **B. Tech – Artificial Intelligence and Data Science** | JCT College of Engineering and Technology | 2022 – 2026 | **CGPA: 8.3**

LEADERSHIP EXPERIENCE: Joint Treasurer, JCT Institutions: Managed finances and coordinated events to ensure accountability and smooth operational workflows.